Basic formulas:

$$a^0 = 1$$

$$a^1 = a$$

$$a^{-n} = \frac{1}{a^n}$$

$$a^{\frac{1}{n}} = \sqrt[n]{a}$$

Basic formulas:

$$a^m \times a^n = a^{m+n}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$(a^m)^n = a^{m \times n}$$

$$a^{m^n} = a^{(m^n)}$$

$$(ab)^n = a^n \times b^n$$

Basic formulas:

If $a^m = a^n$, then

$m = n$ $\quad a \neq 0, 1, -1$

If $a^m = b^m$, then

$a = \pm b$, if m is even

$a = b$, if m is odd

Things to memorize:

- $2^{12}, 3^6, 5^5, 6^4$
- Squares up to 30
- Cubes up to $12, (15, 21)$
- $\sqrt{2}, \sqrt{3}, \sqrt{5}$

Recognize the rational number:

a) $\sqrt[4]{27}$ b) $\sqrt[3]{16}$ c) $\sqrt{11}$ d) $\sqrt[5]{243}$

Which of the following is a surd?

a) $\sqrt[4]{81}$ b) $\sqrt[5]{32}$ c) $\sqrt[6]{729}$ d) $\sqrt[3]{25}$

$$\left[\sqrt[14]{\sqrt[3]{\left(\sqrt[5]{\left(\sqrt[3]{5^6}\right)^2}\right)^{15}}}\right]^7$$

a) 5^2 b) 5^4 c) 5^8 d) 5^{12}

$$\frac{\left((xy)^{a-b}\,(xy)^{b-c}\right)}{(xy)^{a-c}} = ?$$

a) 1 **b) 0** **c) xy** **d) xy^{abc}**

$$\left(\frac{x^a}{x^b}\right)^{a+b} \times \left(\frac{x^b}{x^c}\right)^{b+c} \times \left(\frac{x^c}{x^a}\right)^{c+a} = ?$$

a) 1 **b) 0** **c) x** **d) x^{abc}**

The value of $\left(\frac{x^a}{x^b}\right)^{\frac{1}{ab}} \cdot \left(\frac{x^b}{x^c}\right)^{\frac{1}{bc}} \cdot \left(\frac{x^c}{x^a}\right)^{\frac{1}{ca}} =$

$x^{b^{a^{1}}} \cdot x^{c^{b^{1}}} \cdot x^{a^{c^{1}}}$

a) 1 b) $\frac{1}{x^{abc}}$ c) $\frac{1}{x^{ab+bc+ca}}$ d) None of these

$4^{4x+1} = \frac{1}{64}$, then x is

a) $\frac{1}{2}$ b) -1 c) $-\frac{1}{2}$ d) $-\frac{1}{6}$

If $5^{5x+5} = 1$, then x equals

a) 0 b) -1 c) 1 d) $-\frac{4}{5}$

If $27^{2x-1} = 243^3$, then x equals

a) 3 b) 6 c) 7 d) 9

If $8^{3x-5} = \frac{1}{32^{7-4x}}$ then $x = ?$

$\frac{1}{\quad}$ $x = ?$

a) $\frac{16}{9}$ b) $\frac{20}{11}$ c) $\frac{25}{13}$ d) 2

If $\left(\frac{x}{y}\right)^{5a-3} = \left(\frac{y}{x}\right)^{17-3a}$ then $a = ?$

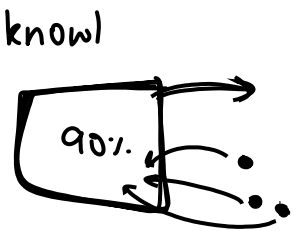
x^{5a-3} y^{17-3a}

a) -5 b) -6 c) -7 d) 7

$x^{x\sqrt{x}} = \left(x\sqrt{x}\right)^{x}$, then x equals

$x^{x\sqrt{x}} = x\sqrt{x}$

a) $\frac{4}{9}$ b) $\frac{2}{3}$ c) $\frac{9}{4}$ d) $\frac{3}{2}$



If $(x^x)^{\frac{5}{4}} = x^{x^{\frac{5}{4}}}$ then $x = ?$

x^5 x^5 $x = ?$

a) $\frac{125}{64}$ b) $\frac{625}{256}$ c) $\frac{25}{16}$ d) $\frac{5}{4}$

Simplify $(5 \times 5 \times 5 \times 5 \times 5)^5 \times (5 \times 5 \times 5)^5 \div 5 = (125)^?$

a) 15 b) 13 c) 21 d) 14

Find the value of $\left(\frac{9^3+9^4+9^5+9^6+9^7}{9^1+9^2+9^3+9^4+9^5}\right)^{\frac{1}{2}}$

$\left(\frac{9^3+9^4+9^5+9^6+9^7}{9^1+9^2+9^3+9^4+9^5}\right)^{1}$

a) 81 **b) 9** **c) 729** **d) 3**

If $\left(3^{33} + 3^{33} + 3^{33}\right)\left(2^{33} + 2^{33}\right) = 6^{x}$, then what is the value of x?

a) 34 b) 35 c) 33 d) 33.5

If $x^{y+Z} = 1, y^{x+z} = 1024\ and\ z^{x+y} = 729$ (x, y and z are natural number), then what is the value of $(z+1)^{y+x+1}$?

a) 6561 b) 10000 c) 4096 d) 14641

If $a = 0.4039$ then $\sqrt{4a^2 - 4a + 1} + 5a =$?

a) 2.2117 b) 2.4039

c) 2.8078 d) 1.8273

If $2 < x < 3$ then $\sqrt{(2-x)^2} + \sqrt{(x-3)^2} = ?$

a) 1 b) -1 c) $2x - 5$ d) $5 - 2x$

If $x = y^a$, $y = z^b$ and $z = x^c$, then the value of abc is

a) 1 b) 2 c) -1 d) 0

If $a^x = b$, $b^y = c$ and $xyz = 1$, then what is the value of c^z?

a) a b) b c) ab d) $\frac{a}{b}$

If $a^x = b^y = c^z$ and $abc = 1$, then the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ will be equal to

a) -1 b) 0 c) 1 d) 3

If $a^{\frac{1}{m}} = b^{\frac{1}{n}} = c^{\frac{1}{p}}$ and $abc = 1$, then $(m + n + p)$ is equal to

a) 0 b) 2 c) 1 d) -2

If $2^x = 3^y = 6^{-z}$ then $\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$ is

$\frac{1}{-} + \frac{1}{-} + \frac{1}{-}$

a) 0 b) 1 c) $\frac{3}{2}$ d) $-\frac{1}{2}$

If $4^x = 6^{-y} = 9^z$ then $\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$ is

$\frac{1}{-} + \frac{1}{-} + \frac{1}{-}$

a) 0 b) $-\frac{1}{x}$ c) $-\frac{1}{y}$ d) $-\frac{1}{z}$

If $2^x = 3^y = 12^z$ then $\left(\frac{1}{z} - \frac{1}{y}\right)$ is

$\frac{1}{\quad} - \frac{1}{\quad}$

a) 0 b) $\frac{1}{x}$ c) $\frac{2}{x}$ d) $\frac{3}{x}$

If $(3.7)^x = (0.037)^y = 10000$, then what is the value of $\frac{1}{x} - \frac{1}{y}$

a) 1 b) 2 c) $\frac{1}{2}$ d) $\frac{1}{4}$

Select the number that will come in place of the question mark (?) in the mathematical statement.

$(0.064)^{123} \div 0.16^{47} \times 0.4^{34} \times 0.4^{29} = (0.4)^{?}$

a) 350 b) 320 c) 338 d) 341

If $A = \frac{\sqrt{0.0004} \times \sqrt[3]{0.000008}}{\sqrt[4]{16000} \times \sqrt[3]{125000} \times \sqrt[4]{810}}$ and $B = \frac{\sqrt[3]{0.729} \times \sqrt[4]{0.0016}}{\sqrt{0.16}}$, then what is $A \times B$?

$0.0004 \times^{3} 0.000008$

$16000 \times \ 125000 \times \ 810$

$B = {}^{3} 0.729 \times^{4} 0.0016$

a) 7×10^{-7}

b) $\left(\frac{7}{4}\right) \times 10^{-8}$

c) 5×10^{-8}

d) $\left(\frac{7}{3}\right) \times 10^{-7}$

$$\frac{.02 \times .02 \times .9 \times .2}{2 \times 3 \times 10 \times 5 \times 10 \times .4}$$

Solve $\sqrt{2\sqrt{2}}$

Solve $\sqrt{2\sqrt{2\sqrt{2}}}$

Solve $\sqrt{6\sqrt{2\sqrt{3}}}$

Solve $\sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}}$

Solve $\sqrt{4\sqrt{8\sqrt{16}}}$